AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 25 · STUDENT LESSON

Metrics Match Consequences

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Name the decision. Count the errors. Check the capacity.

Choose a metric suite under declared consequences and bounded development evidence.

Keep one habit: Complete threshold fixture → pictured policy swap → three stakeholder council requests → feasibility check → independent fresh costs.

MAIN LESSON · SECTION 1

What decision will this metric support?

> State error consequences and review capacity before choosing the most attractive number.

Learning goals

I can apply a declared **score threshold** including equality.

I can compare **false alarm**s and misses at two **threshold**s.

I can separate validation choices from final testing.

Retrieve the recall denominator

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a **score** for your ability.
- Without opening the old lesson, explain one key idea from Train, Validation, Test. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Train a Regression Model. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.
- Which denominator defines recall?

End goal

By the end, I can explain metrics match **consequence**s using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Metric choice depends on intended use, the positive class, who is affected, and the **consequence**s of each error.



LEARNING TARGET A reviewer can inspect at most 10 alerts daily. Model A flags 20 cases with higher recall; B flags 8 with lower recall. What **evidence** is needed before choosing a **threshold**?

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Declare positive,threshold direction,cost and data role

> A consequence-aware choice needs an explicit decision rule.

Core threshold explanation and fictional scope

A classifier **score** is not automatically a probability. Declare blocked positive and predict blocked when **score** is at least the **threshold**. Changing this boundary changes decisions without necessarily re**training** the model.

Select it using development **evidence** and a task-relevant rule, not final-test answers.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **threshold**s and **consequence**s.

Complete current cost contract

For the main six-record exercise, blocked is positive, predict blocked when score>=threshold, and use fictional cost 1 point per false alarm plus 4 per miss. Compare only thresholds 0.5 and0.3on the same supplied development records. These points are invented teaching priorities, not measured real-world harms. The separate leaf picture uses different costs and four records; keep its values separate.

A ranking score is not automatically a probability.

▼ Earlier wording, alternatives and picture transcripts

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use metrics match **consequence**s to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Trace all six decisions at both declared thresholds

> The equality case changes the false-alarm count.

Complete six-record fixture and counts

Six records have **score**/reference pairs A .9/blocked, B .7/clear, C .6/blocked, D .4/blocked, E .3/clear, F .1/clear. At **threshold** .5, TP=2, FP=1, FN=1, TN=2. At .3, TP=3, FP=2, FN=0, TN=1.

Equality is positive, so E is flagged at .3.

Optional complete cost calculation script

```
scores = [0.9, 0.7, 0.6, 0.4, 0.3, 0.1]
actual = [True, False, True, True, False, False]
for threshold in [0.5, 0.3]:
    predicted = [s >= threshold for s in scores]
    fp = sum(p and not a for p, a in zip(predicted, actual))
    fn = sum(not p and a for p, a in zip(predicted, actual))
    print(threshold, fp, fn, fp + 4 * fn)
# Output rows: 0.5 1 1 5, then 0.3 2 0 2.
```

Predict before running

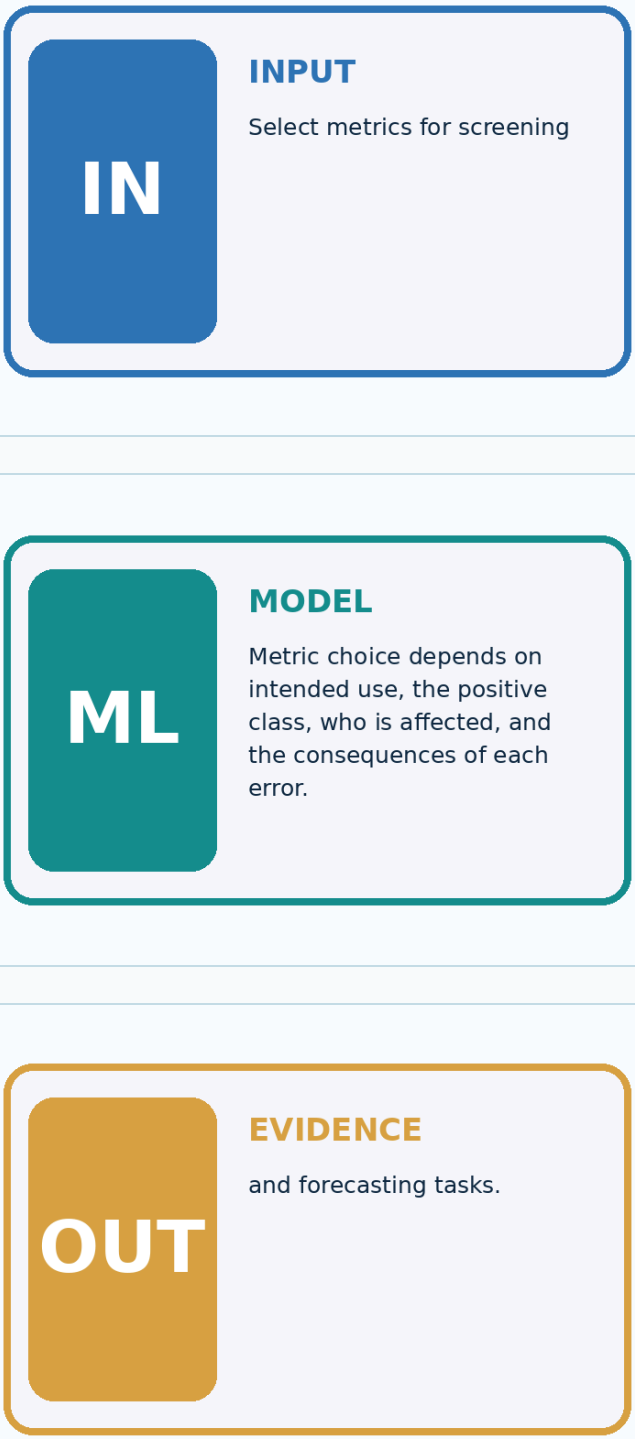
Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

▼ **Earlier wording, alternatives and picture transcripts**

ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.



QUESTION Select metrics for screening, recommendation, and forecasting tasks.

MODEL IDEA Metric choice depends on intended use, the positive class, who is affected, and the **consequence**s of each error.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Screening task treats missed positives as costly, so emphasize recall and human follow-up. Recommendation task may track precision among shown items plus user benefit.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

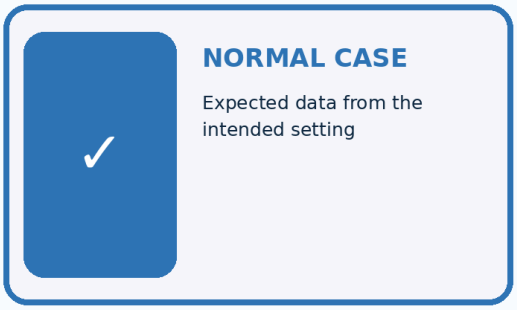
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

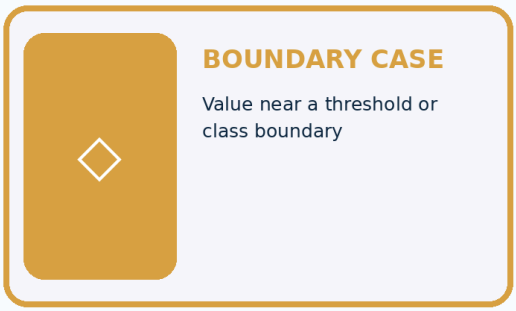
Compare the same metric under different priorities

> The preferred threshold can change when costs or capacity change.

Four evaluation conditions



NORMAL Expected data and expected behavior.

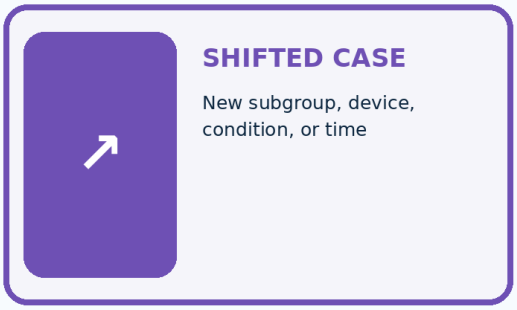


Keep this boundary clear

BOUNDARY Value close to a **threshold**, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: Miss and false-alarm costs swap to1 and 5.

▼ **Check after predicting**

Prediction: **Threshold**0.7 costs1 while0.4 costs5; preferred **threshold** switches.

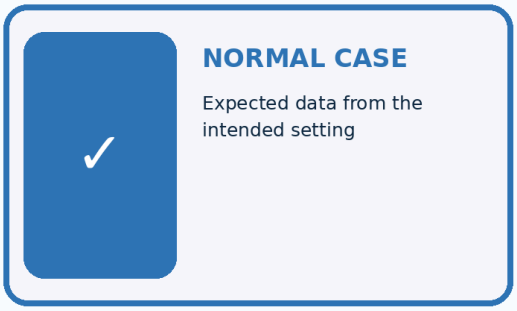
Reason: Metric choice encodes the declared **consequence**s, not a universal best **threshold**.

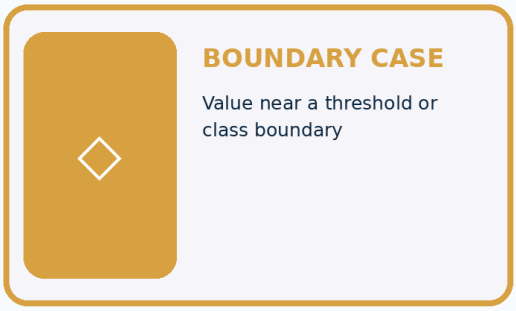
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

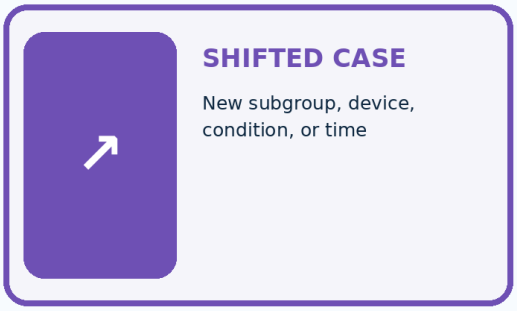
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a **threshold**, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read predictions,costs and policy together

> Changing a consequence weight does not change the supplied reference labels.

Complete pictured scores and separate-policy scope

Translate a **threshold** choice into missed-leaf and false-alarm costs.

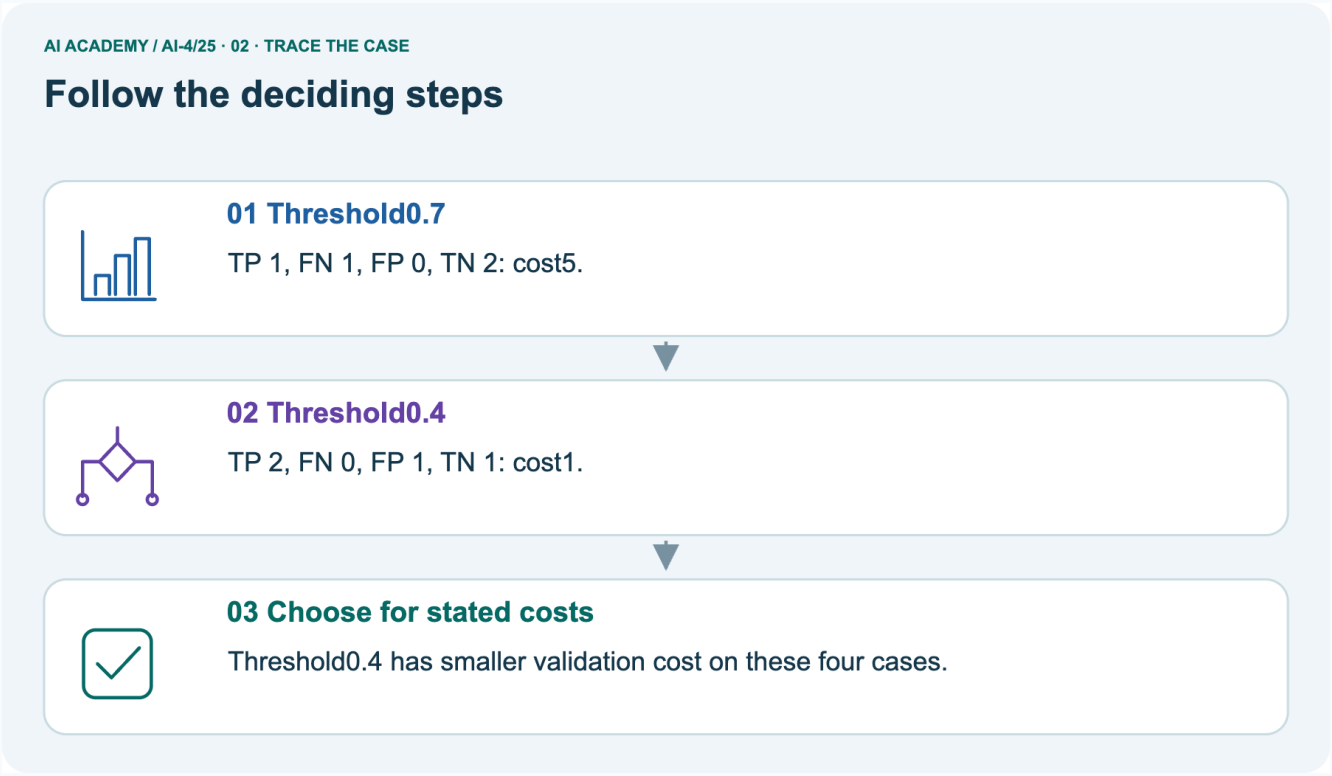
Scores/reference: 0.9 spotted, 0.6 healthy, 0.4 spotted, 0.1 healthy. Predict spotted at **score**>=**threshold**. Miss cost5; false-alarm cost1.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Main A–Fthresholds0.5/0.3 both classify 4/6 correct, but costs under FP 1/FN 4 are5 and2. Reversing main priorities to FP 4/FN 1 gives5 and8. Picture instead has four leaf scores 0.9 spotted, 0.6 healthy, 0.4 spotted, 0.1 healthy; thresholds 0.7/0.4 cost5 and1 under FP 1/FN 5. Swapping the pictured costs to FP 5/FN 1 gives1 and5.

These are separate fixtures and separate policy changes; no single threshold is universally best or validated for real leaf-health decisions.

Trace both pictured thresholds and costs



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Threshold0.7	TP 1, FN 1, FP 0, TN 2: cost5.
2. Threshold0.4	TP 2, FN 0, FP 1, TN 1: cost1.
3. Choose for stated costs	Threshold0.4 has smaller validation cost on these four cases.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

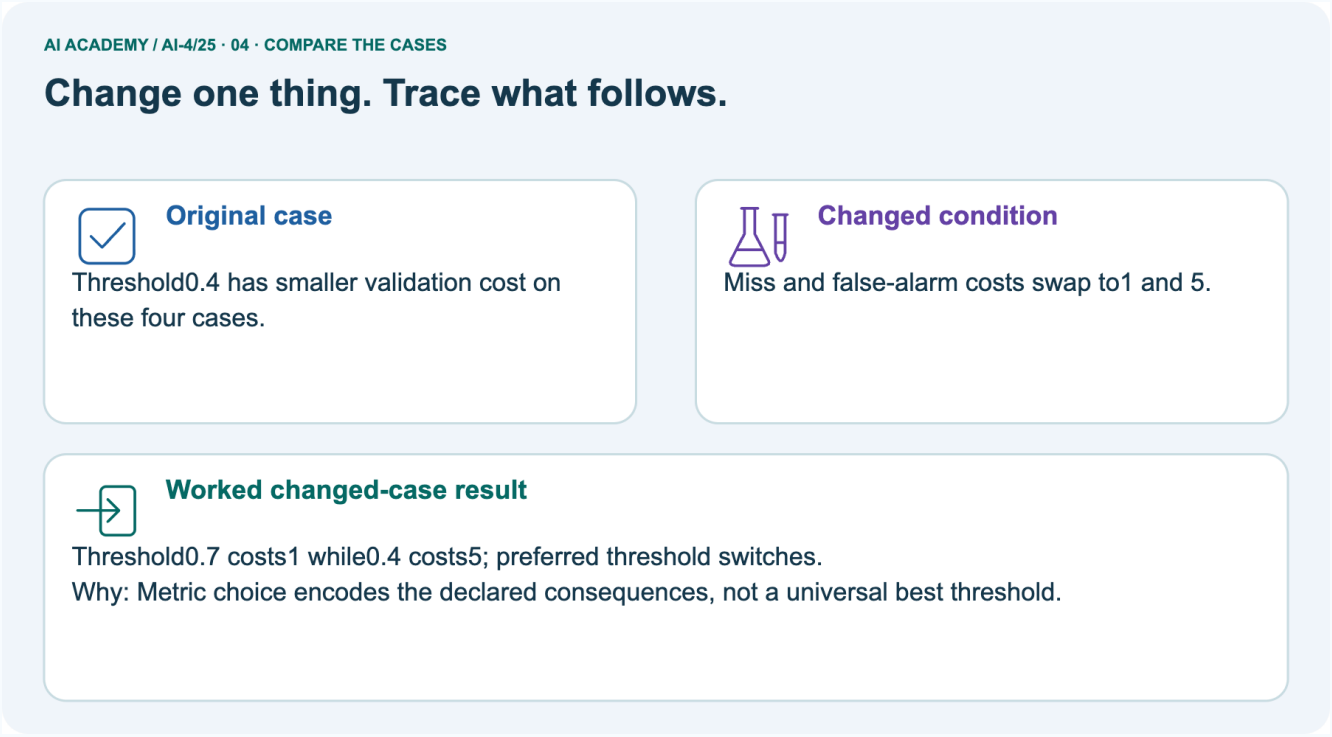
Read the picture in order. **Threshold**0.4 has smaller validation cost on these four cases. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict swapping the pictured consequence weights

- Change one thing: Miss and false-alarm costs swap to1 and 5. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain real-use and reference-label limits

- Toy costs illustrate decision design; real use needs agreed **consequence**s, review capacity and fresh evaluation.

- Read the labelled picture: Which labels determine the false-alarm count at **threshold** 0.4? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to the main six-record counts

Thresholds and consequences at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Threshold .5	One false alarm one miss	Four of six correct
Threshold .3	Two false alarms no misses	Four of six correct
Decision rule	Include equality	Record before evaluation

Lowering the threshold here finds D but adds E as a false alarm. The two accuracies are equal while consequences differ. Precision is 2/3 at .5 and 3/5 at .3; recall rises from 2/3 to 1. This behavior follows these fixed scores and records, not a promise of future performance.

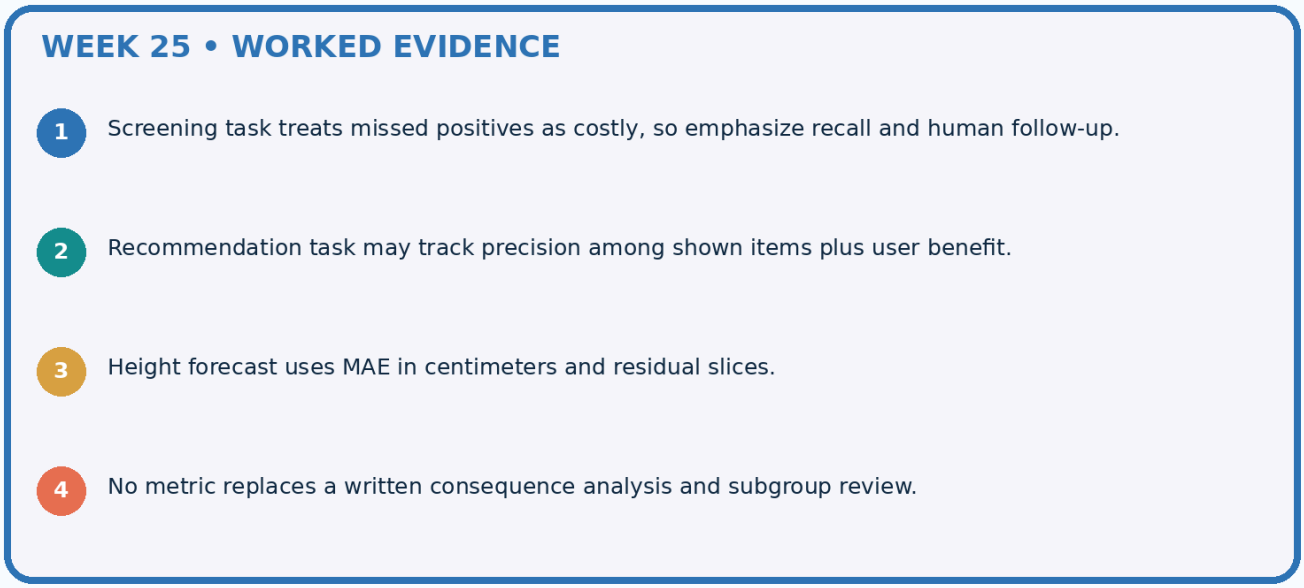
Give TP, FP, FN, TN at both thresholds. Explain the steps or evidence behind your response.

▼ **Earlier wording, alternatives and picture transcripts**

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.



STEP 1 Screening task treats missed positives as costly, so emphasize recall and human follow-up.

STEP 2 Recommendation task may track precision among shown items plus user benefit.

STEP 3 Height forecast uses MAE in centimeters and residual slices.

STEP 4 No metric replaces a written **consequence** analysis and subgroup review.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a **score**.

Recalculate, retrace, or restate the **evidence** independently:

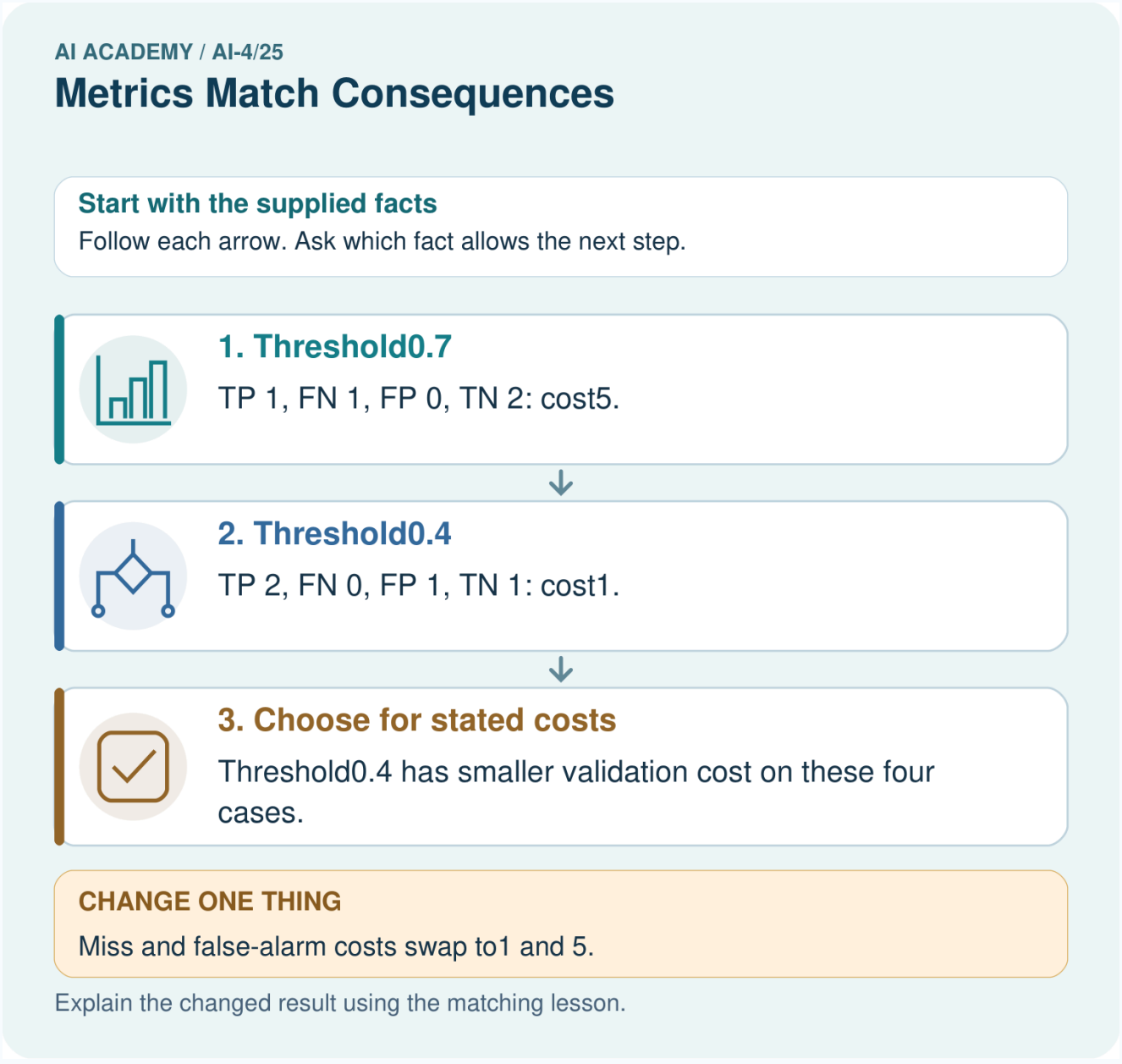
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Metrics Match Consequences

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a metric treated as a complete decision policy

> Counts,consequences and human capacity answer different questions.

Diagnose the claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE The same highest-accuracy metric should choose every model.

Explain interpretation and next-check limits

- These invented costs support arithmetic practice, not real robot deployment. Stakeholders must justify actual **consequence**s.

  Record every tried **threshold** and retain a separate final test; a **threshold** is part of the selected method.

> **Why not select the threshold with the lowest final-test cost?**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: The same highest-accuracy metric should choose every model. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

MAIN LESSON · SECTION 8

Practise the unchanged predictions under two cost rules

> The evaluation evidence stays fixed while the declared priorities differ.

Try supported W2

Use workbook W2 for the connected example “**Threshold**s and **consequence**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Trace main costs5/2and reversed5/8

Suppose the classroom cost is one point per **false alarm** and four per miss. Total costs are 5 at .5 and 2 at .3, so .3 wins among these two candidates under this rule.

If **false alarm**s cost four and misses cost one, costs are 5 and 8, reversing the choice.

Explain which evidence supports the choice

- Use the worked case for **Threshold**s and **consequence**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture’s false-alarm references

- Which labels determine the false-alarm count at **threshold** 0.4? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the fresh four-record independent task

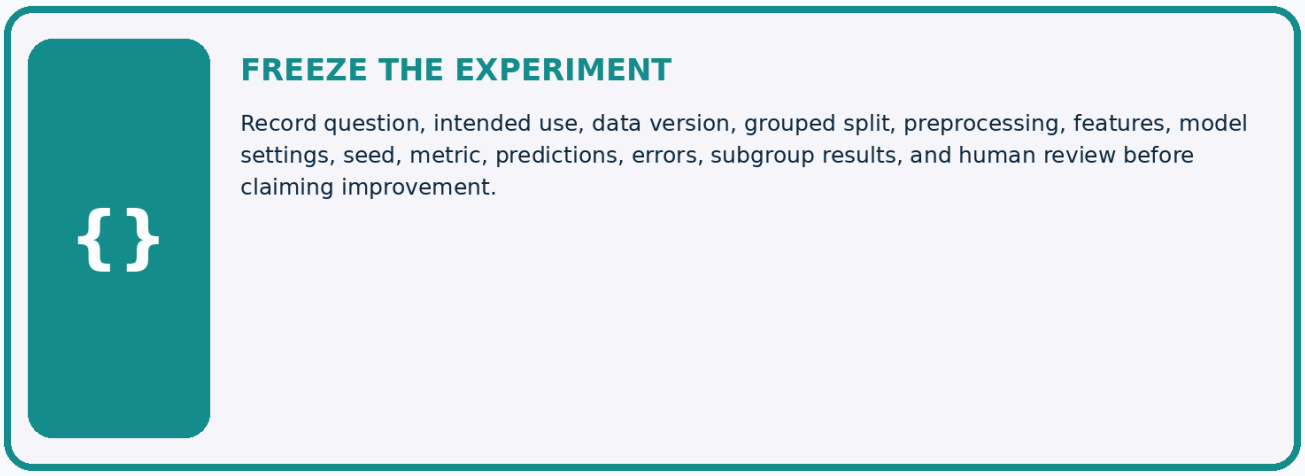
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success **threshold**

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Count the consequences

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete three-scenario metric council

> Every score,reference,outcome and fictional capacity needed for the comparison is supplied.

Complete six-case decision record

Main case ID · Score · Reference · Prediction at0.5 · Prediction at0.3

> **Main case ID:** A
> **Score:** 0.9
> **Reference:** blocked
> **Prediction at0.5:** blocked
> **Prediction at0.3:** blocked

> **Main case ID:** B
> **Score:** 0.7
> **Reference:** clear
> **Prediction at0.5:** blocked
> **Prediction at0.3:** blocked

> **Main case ID:** C
> **Score:** 0.6

Reference: blocked

Prediction at0.5: blocked

Prediction at0.3: blocked

Main case ID: D

Score: 0.4

Reference: blocked

Prediction at0.5: clear

Prediction at0.3: blocked

Main case ID: E

Score: 0.3

Reference: clear

Prediction at0.5: clear

Prediction at0.3: blocked

Main case ID: F

Score: 0.1

Reference: clear

Prediction at0.5: clear

Prediction at0.3: clear

Three complete council requests

Complete fictional Metric Council request · Available evidence and requirement

Complete fictional Metric Council request: 1 Marked-lane review queue for this six-case development batch

> **Available evidence and requirement:** Use A–Fabove; FPcost1, FNcost4; hard capacity 3 alerts per batch. All six references observed. Choose only among 0.5/0.3.

> **Complete fictional Metric Council request:** 2 Three-slot classroom book recommendations
>
> **Available evidence and requirement:** Shown itemsR1, R2, R3 have reviewed relevance labels relevant, not relevant, relevant. There are 2 other relevant unshown items in the complete 5-item relevant/not-relevant pool. Compare shown-item precision and found-relevant recall; no reader-benefit observations supplied.

> **Complete fictional Metric Council request:** 3 Day-7 fictional plant-height forecast
>
> **Available evidence and requirement:** Matched actual heights 10, 12, 14 cm; supplied predictions 9, 14, 14 cm. Need an error-size report in cm and the largest miss; no classification labels or user-cost weights supplied.

Propose the metric suite and bounded decision

1. Metric Council: first count the complete A–Fpredictions, then address all three supplied stakeholder requests. Name the positive/target definition, each denominator, the cost or capacity constraint and the evidence the metric cannot show. For the hard-capacity case, check feasibility before minimizing cost. Keep threshold selection as development evidence and do not claim final performance.

   Use the reasoned guidance only after a first proposal. Keep the independent four-score task separate for an uncoached attempt.

▼ Check reasoned council guidance after trying

Council case · Reasoned metric/constraint decision

> **Council case:** 1 Lane batch

Reasoned metric/constraint decision: At0.5:3 alerts, TP 2/FP 1/FN 1/TN 2, cost 5, precision 2/3, recall 2/3. At0.3:5 alerts, TP 3/FP 2/FN 0/TN 1, cost 2, precision 3/5, recall 1. Hard capacity 3 makes0.3 infeasible under this proposal despite lower fictional error cost. Choose0.5 among supplied feasible options or redesign staffing/workflow and reassess; do not silently discard two alerts.

Council case: 2 Book slots

Reasoned metric/constraint decision: Precision among shown 2/3; there are 4 relevant items total, so shown relevant recall 2/4. These do not establish reading enjoyment, learning benefit or a fair display policy; ask for those outcomes/context.

Council case: 3 Height

Reasoned metric/constraint decision: Residuals+1, -2, 0 cm; absolute errors 1, 2, 0; MAE 1 cm, largest 2 cm. Keep case pairing and units; classification accuracy/precision do not answer this numeric-error question.

Normal,equality,missing-policy and changed-cost probes

Complete consequence probe	Expected bounded check
Normal main cost comparison without capacity constraint	0.3 cost2 versus0.5 cost5 under FP 1/FN 4.
Boundary main scoreE=0.3at threshold 0.3	Included as positive, false alarm because referenceclear.
Missing stakeholder weights or actual review capacity	State unknowns and request the decision information; do not invent an authorized universal winner.
Changed main cost policy FP 4/FN 1	Costs0.5→5, 0.3→8; the cost-based preference reverses on these same records.

Keep evidence and execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Boundary
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Missing
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Shifted
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

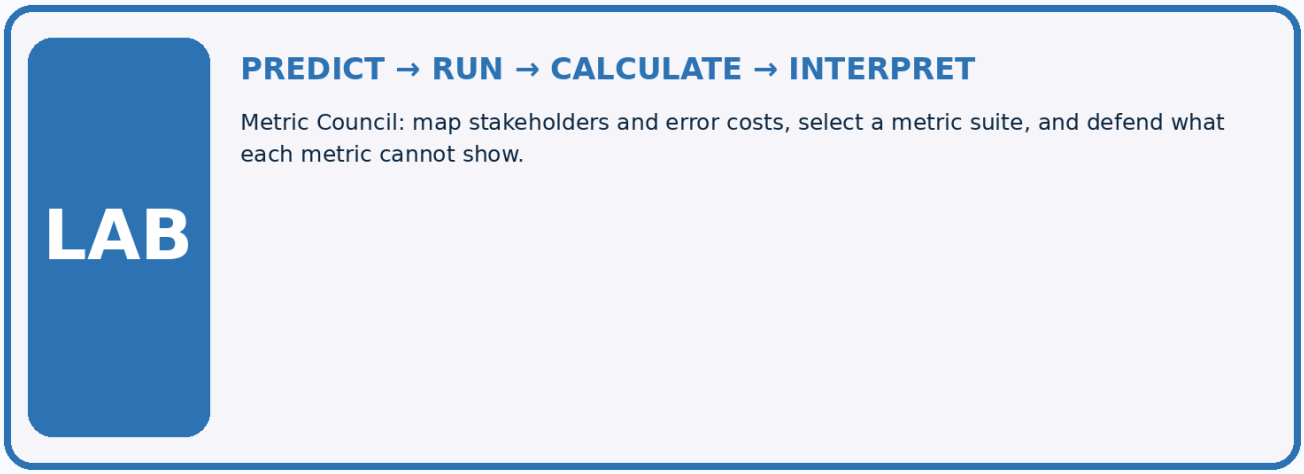
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Metric Council: map stakeholders and error costs, select a **metric suite**, and defend what each metric cannot show.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare new thresholds under an explicit cost rule

> Count all four outcomes before applying the consequence weights.

Complete independent scores and cost scope

Scores/references are .8/blocked,.5/clear,.2/blocked,.1/clear. Compare **threshold**s .5 and .2 using the same cost.

For this new four-record task, “same cost”means the main rule:false alarm 1 point, miss 4 points. Use blocked positive and score>=threshold, including equality at0.5 and0.2. No review-capacity constraint is supplied for this task; do not import the separate council batch capacity. Report the preference among these two thresholds under the stated fictional costs without a final-test or real-world claim.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply independently

Week 25 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Missing a diseased plant is considered more costly than a **false alarm**, but staff can review only ten alerts daily. Which evaluation trade-off should be discussed?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Approve the Plant Classifier **metric suite**: accuracy for context, spotted precision/recall, confusion matrix, and subgroup slices.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

A reviewer can inspect at most 10 alerts daily. Model A flags 20 cases with higher recall; B flags 8 with lower recall. What **evidence** is needed before choosing a **threshold**?

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project metric suite tied to its decision

> A human-capacity constraint should be visible alongside the error tradeoff.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Approve the Plant Classifier **metric suite**: accuracy for context, spotted precision/recall, confusion matrix, and subgroup slices.

Keep accuracy for context, the positive-class matrix, precision/recall and relevant slices with denominators. Record which stakeholder consequence each metric addresses, fictional versus agreed actual cost weights, review capacity, threshold and equality policy, and who makes the final decision.

A probability interpretation needs its own evidence; no metric suite by itself establishes benefit or authorizes consequential use. Preserve source and development/final roles.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits,uncertainty and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

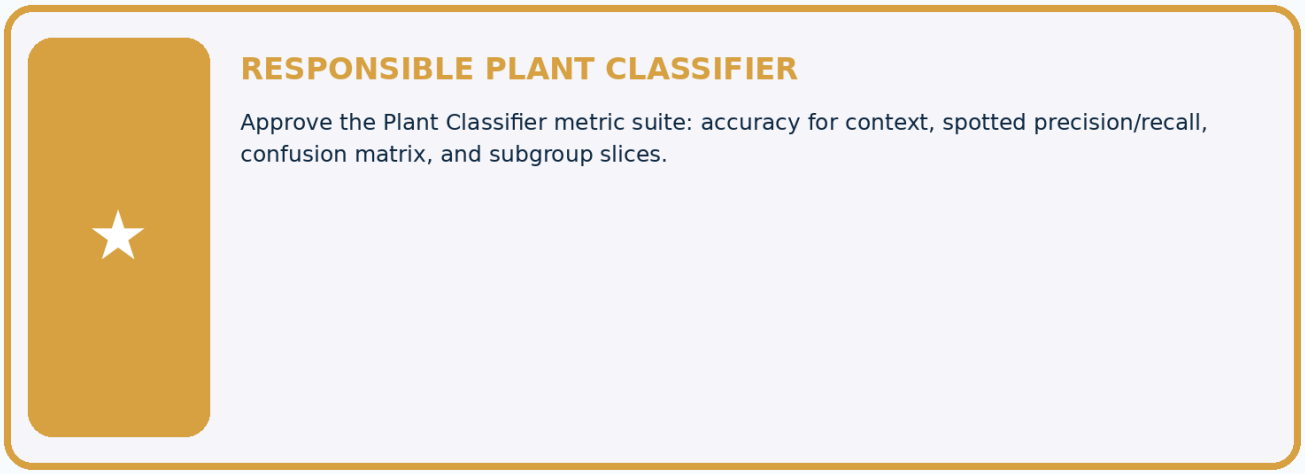
My **evidence**-linked project decision:

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish with the metric’s question and the decision it leaves open

> Explain how capacity or consequence weights can change the recommendation.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Apply a declared **score threshold** including equality

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
consequence	real effect of a prediction error	
threshold	score boundary for a class decision	
cost matrix	relative impact assigned to error types	
metric suite	multiple measures covering distinct concerns	

Closer-explanation terms

Word	Meaning
Score	A model output used to rank or decide
Threshold	A boundary that converts a score to a label
False alarm	A positive prediction for a negative reference

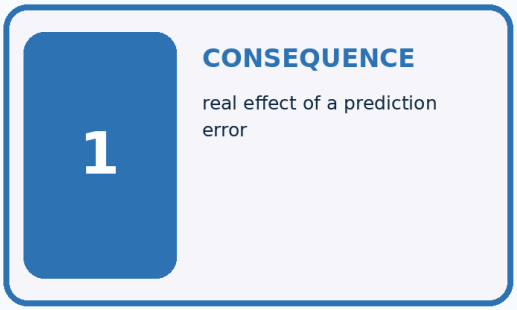
Word	Meaning
Cost rule	A declared way to value different mistakes

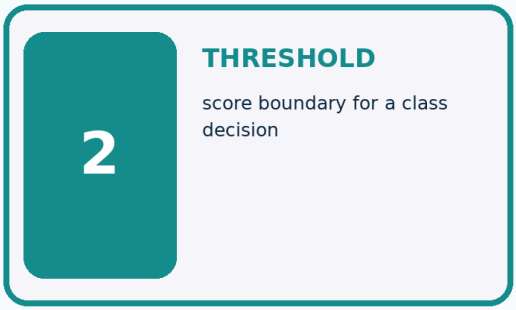
▼ Earlier wording, alternatives and picture transcripts

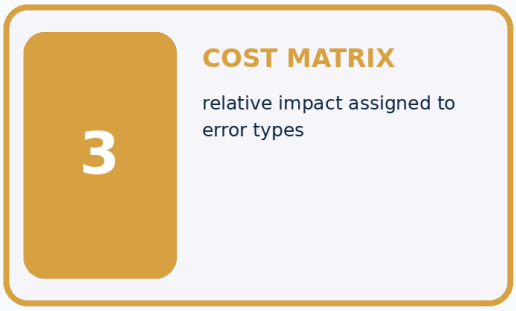
VOCABULARY

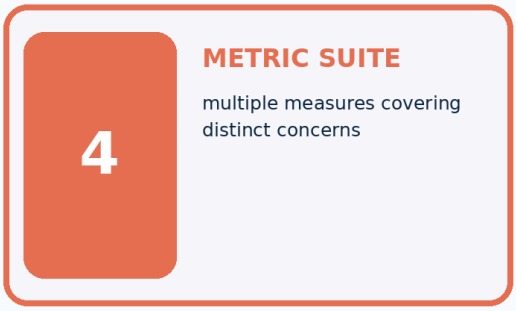
Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR The same highest-accuracy metric should choose every model.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **consequence**, **threshold**, **cost matrix**, **metric suite**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE The same highest-accuracy metric should choose every model.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

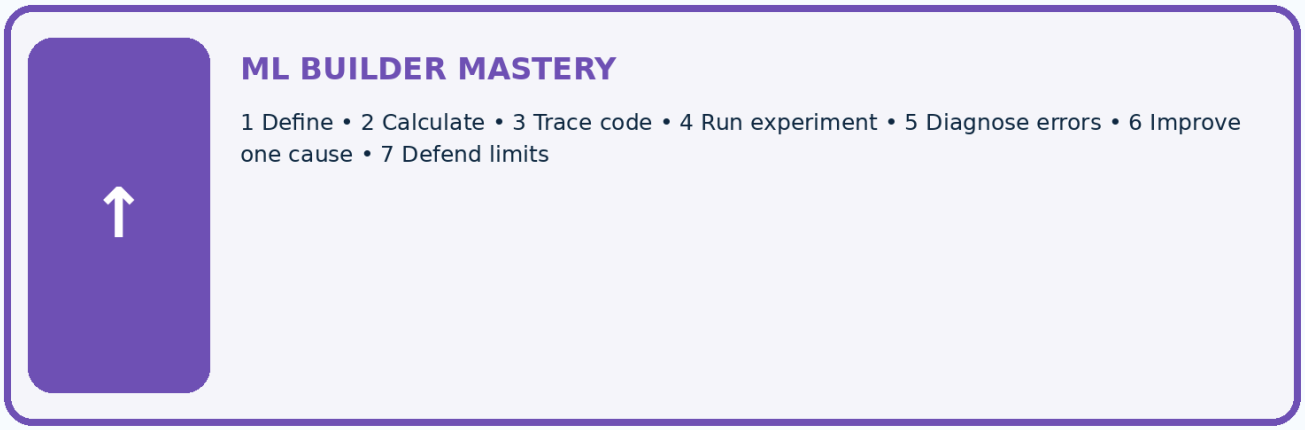
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected fresh four-score threshold-and-cost task for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

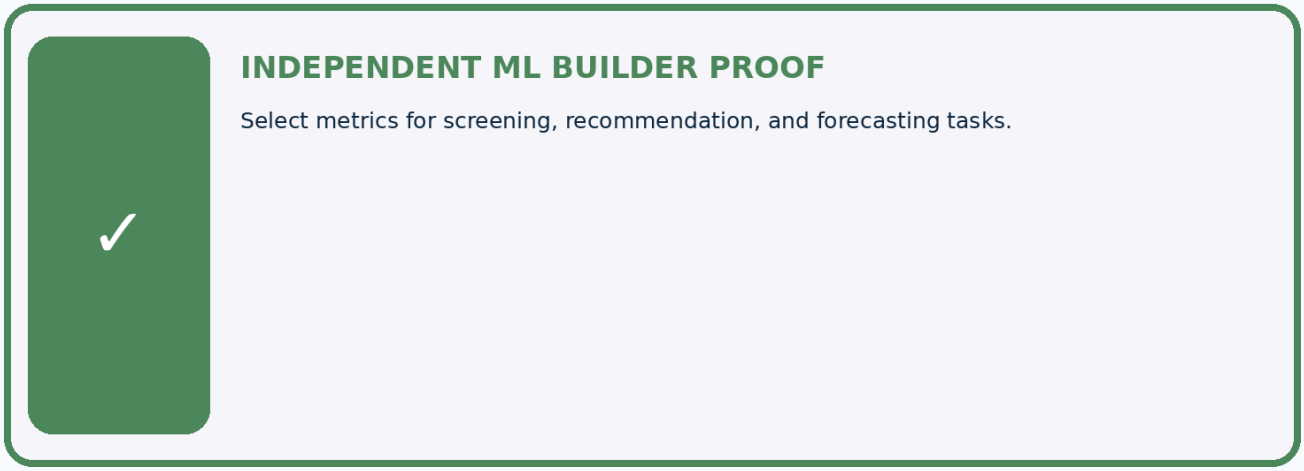
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Select metrics for screening, recommendation, and forecasting tasks.

NEXT BRIDGE Carry the experiment **evidence** into: Reproducible Experiments.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Spotting alerts can miss disease signs or create unnecessary healthy-leaf reviews.

Connect metrics to **consequence**s.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Recall measures finding positives; precision measures alert correctness. Set tradeoffs using **consequence**s, review capacity and validation **evidence**.

Evidence or stage	Value or result
False negative	Spotted leaf missed
False positive	Healthy leaf creates an alert
Recall	Fraction of positives found
Precision	Fraction of alerts correct

▼ Earlier wording, alternatives and picture transcripts

Week 25 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Connect error types to consequences

Spotting alerts can miss disease signs or create unnecessary healthy-leaf reviews.

Connect metrics to **consequence**s.

Recall measures finding positives; precision measures alert correctness. Set tradeoffs using **consequence**s, review capacity and validation **evidence**.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Apply a declared score threshold including equality	Explain the deciding step, show a trace or test, and state one limit.
Compare false alarms and misses at two thresholds	Explain the deciding step, show a trace or test, and state one limit.
Separate validation choices from final testing	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Translate a **threshold** choice into missed-leaf and false-alarm costs.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Translate a **threshold** choice into missed-leaf and false-alarm costs.

- Which labelled step matches this detail? **Threshold**0.4 has smaller validation cost on these four cases.
- 1. **Threshold**0.4
- 2. Choose for stated costs
- 3. **Threshold**0.7
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Choose for stated costs **Threshold**0.4 has smaller validation cost on these four cases. Read the picture in order.

Threshold0.4 has smaller validation cost on these four cases. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Miss and false-alarm costs swap to1 and 5. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.

- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. **Threshold**0.7 costs1 while0.4 costs5; preferred **threshold** switches.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: **Threshold**0.7 costs1 while0.4 costs5; preferred **threshold** switches. Metric choice encodes the declared **consequence**s, not a universal best **threshold**.

Repair and keep the record

Review Train, Validation, Test (ai-4/24). Goal: Select a model using a declared validation rule. Short practice: Why does B's zero **training** MAE not settle the original choice?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train a Regression Model (ai-4/22). Goal: Fit a line from a declared **training** candidate set. Short practice: May day3's evaluation label choose the candidate line?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Regression and Residuals (ai-4/19). Goal: Distinguish numeric **prediction** from classification. Short practice: Which statistic reveals the size of errors without cancellation?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work
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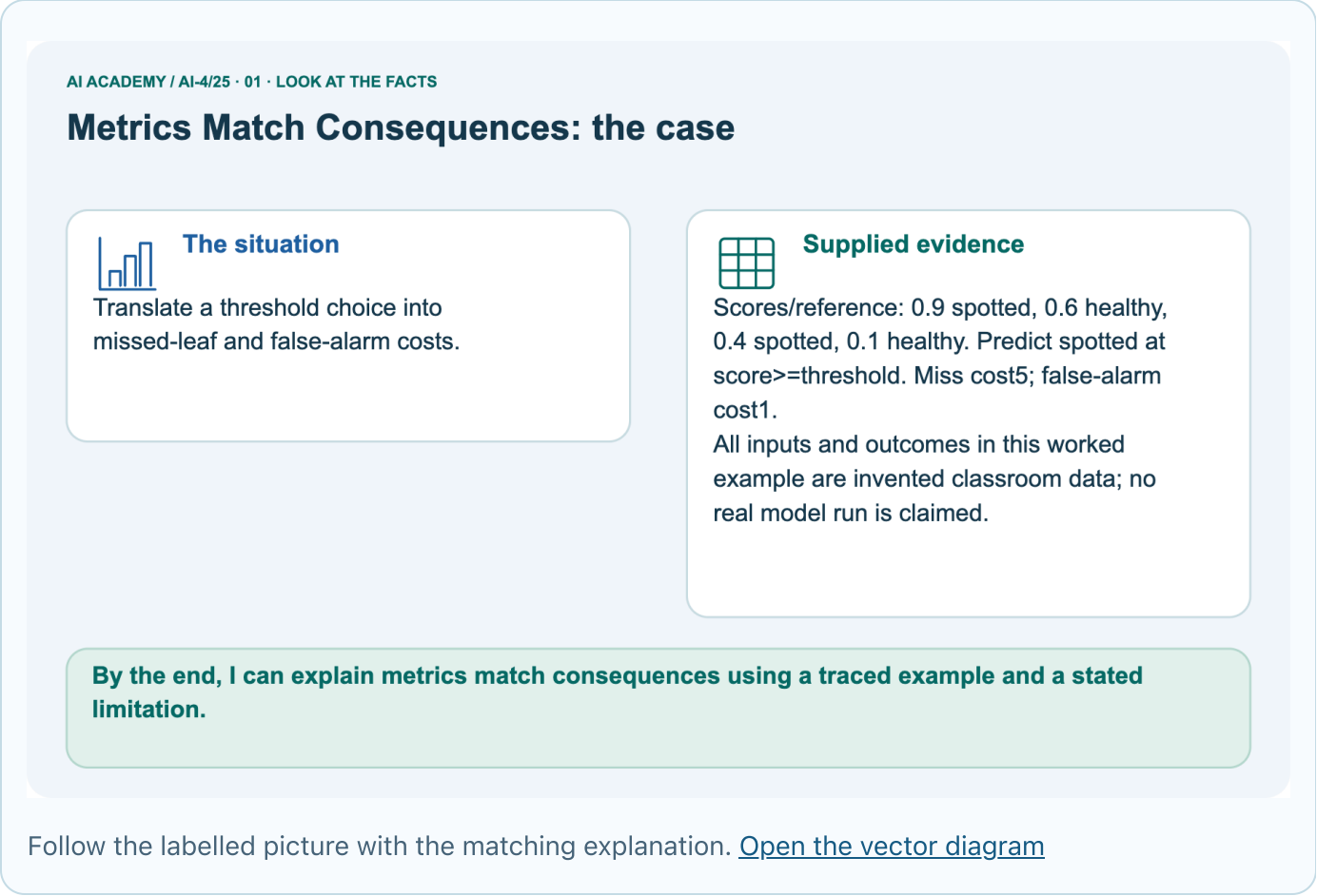
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<summary>Reference · The six-picture journey</summary>
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REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Metrics Match Consequences: the case



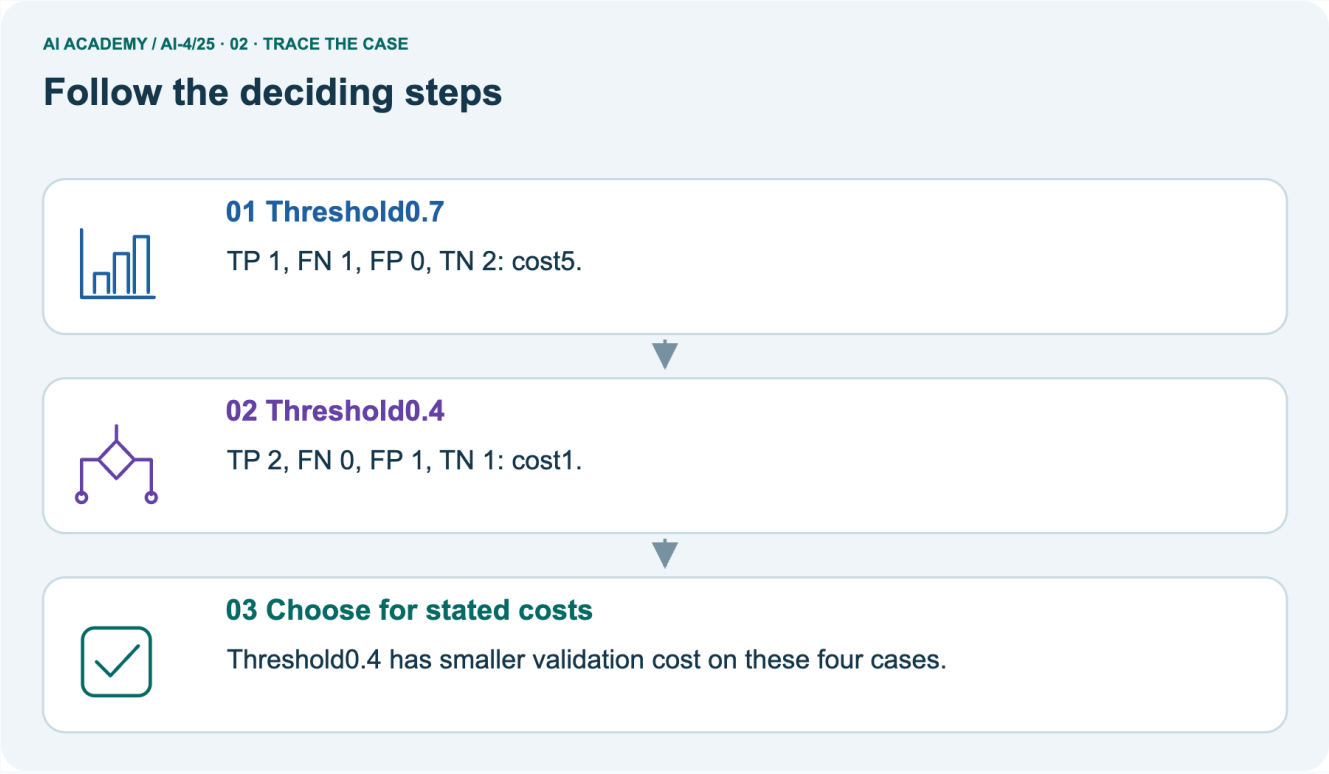
Follow the labelled picture with the matching explanation. Open the vector diagram

Translate a **threshold** choice into missed-leaf and false-alarm costs. Supplied facts: **Score**s/reference: 0.9 spotted, 0.6 healthy, 0.4 spotted, 0.1 healthy. Predict spotted at **score**>=**threshold**. Miss cost5; false-alarm cost1.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

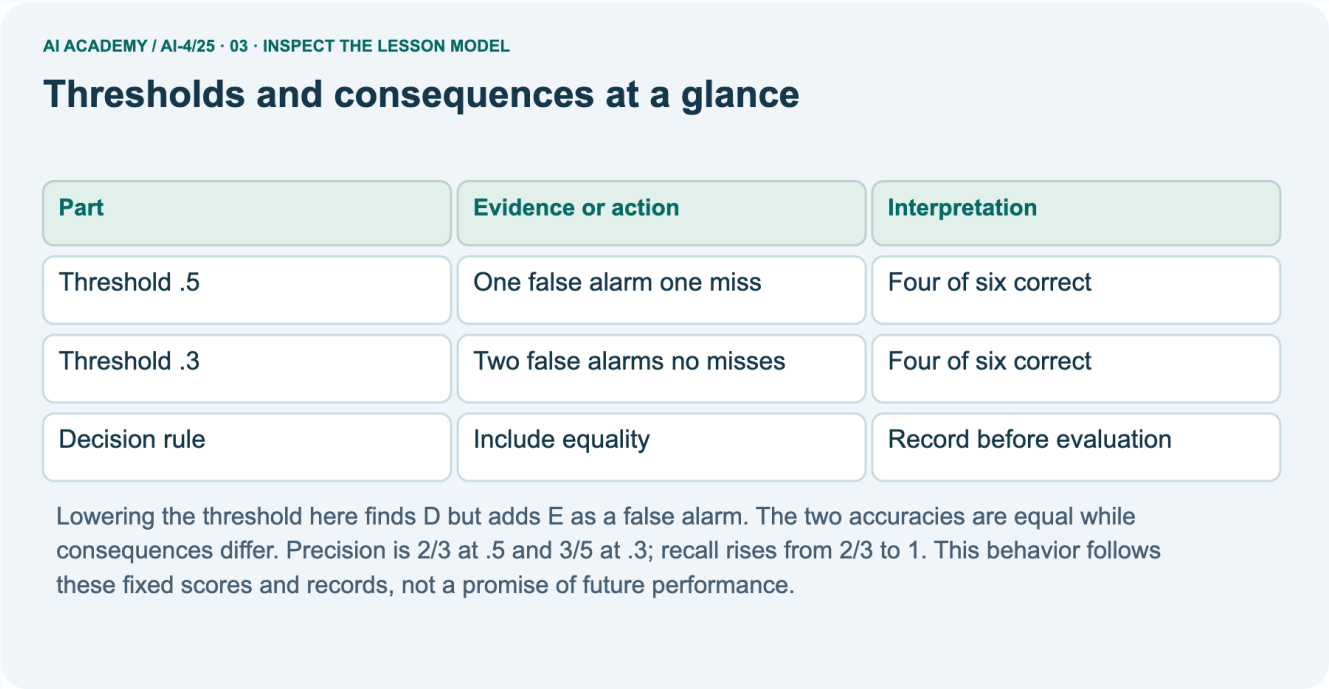


Follow the labelled picture with the matching explanation. Open the vector diagram

Threshold0.7: TP 1, FN 1, FP 0, TN 2: cost5. → **Threshold**0.4: TP 2, FN 0, FP 1, TN 1: cost1. → Choose for stated costs: **Threshold**0.4 has smaller validation cost on these four cases.

Point to each numbered stage and explain which recorded **input** it uses.

03. Thresholds and consequences at a glance



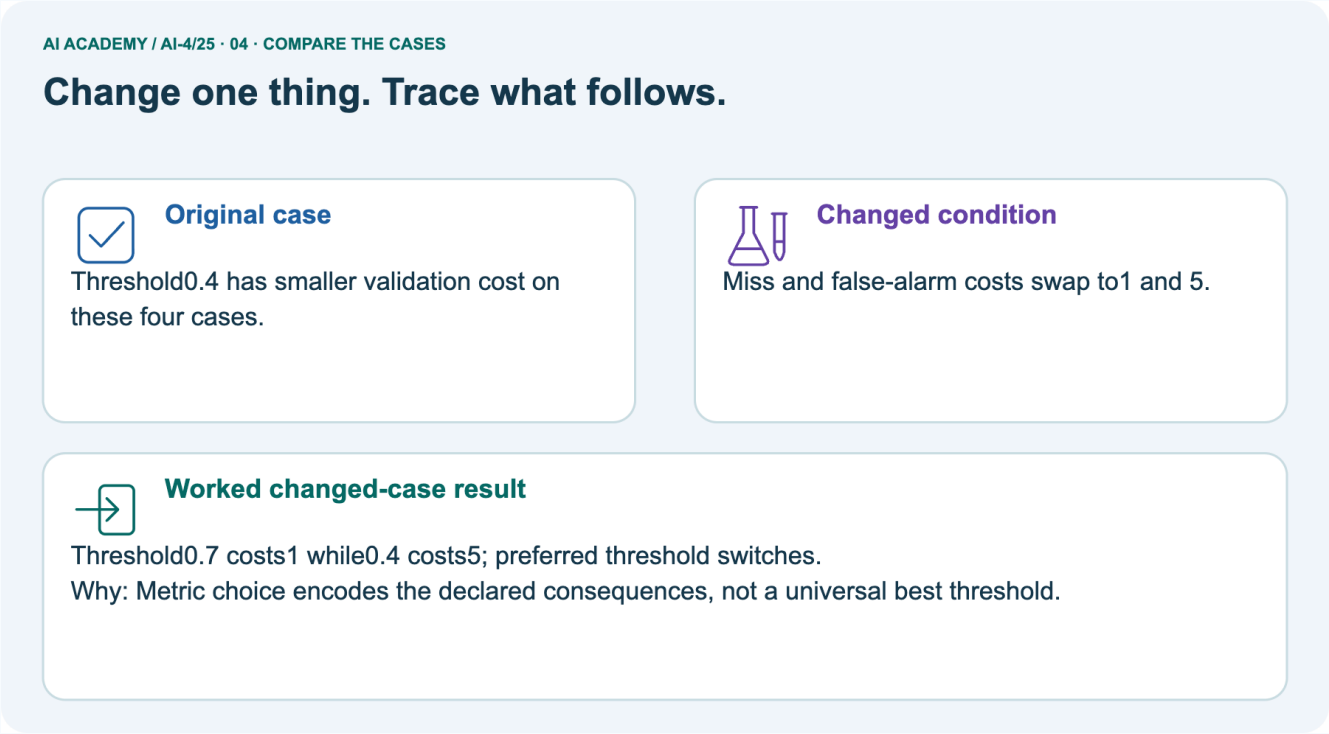
Follow the labelled picture with the matching explanation. Open the vector diagram

Thresholds and **consequence**s at a glance. **Threshold** .5 | One **false alarm** one miss | Four of six correct. **Threshold** .3 | Two **false alarm**s no misses | Four of six correct.

Decision rule | Include equality | Record before evaluation

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



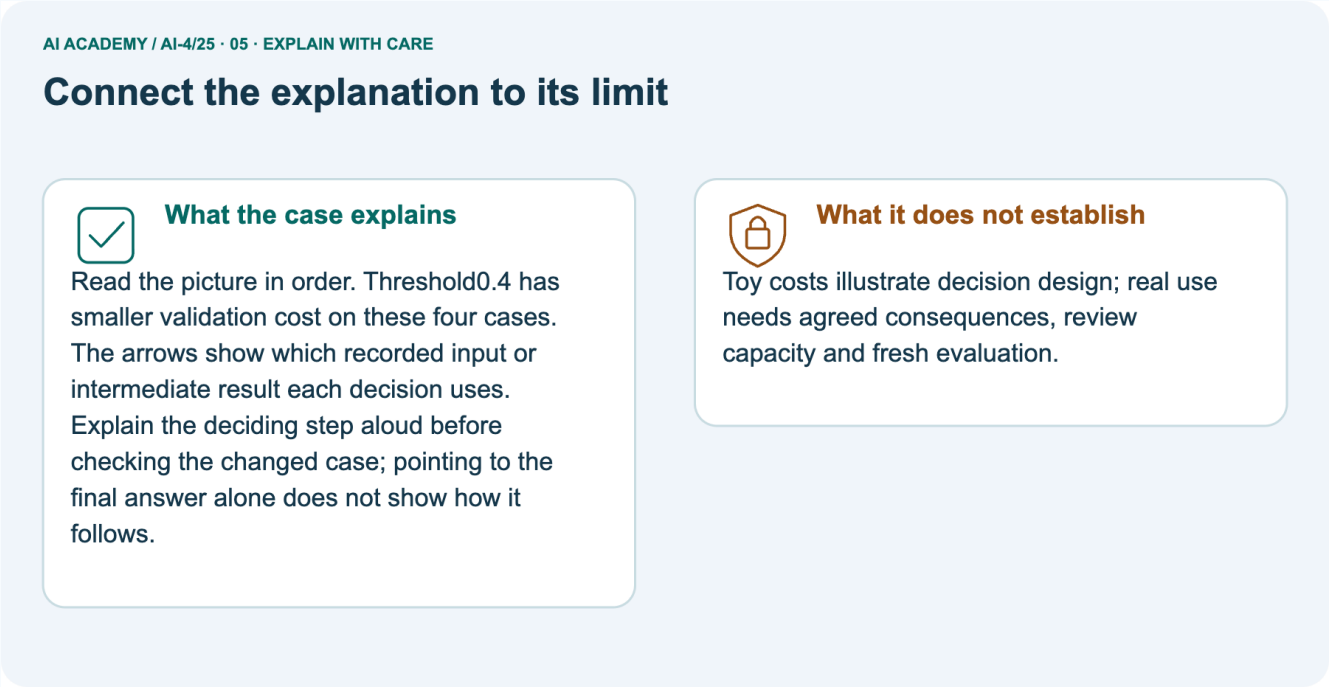
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Threshold**0.4 has smaller validation cost on these four cases. Changed condition: Miss and false-alarm costs swap to1 and 5. Expected result: **Threshold**0.7 costs1 while0.4 costs5; preferred **threshold** switches.

Reason: Metric choice encodes the declared **consequence**s, not a universal best **threshold**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. **Threshold**0.4 has smaller validation cost on these four cases. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Toy costs illustrate decision design; real use needs agreed **consequence**s, review capacity and fresh evaluation.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which labels determine the false-alarm count at **threshold** 0.4? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Metrics Match Consequences: the case

02. Follow the deciding steps

03. Thresholds and consequences at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.